

Problem E. Inversion Count

Time Limit	3599 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Let $A[0 \dots n - 1]$ be an array of n distinct positive integers. If $i < j$ and $A[i] > A[j]$ then the pair (i, j) is called an inversion of A . Given n and an array A your task is to find the number of inversions of A .

Input

The first line contains t , the number of testcases followed by a blank space. Each of the t tests start with a number n ($n \leq 200000$). Then $n + 1$ lines follow. In the i th line a number $A[i - 1]$ is given ($A[i - 1] \leq 10^7$). The $(n + 1)$ th line is a blank space.

Output

For every test output one line giving the number of inversions of A .

Example

Input	Output
2 3 3 1 2 5 2 3 8 6 1	2 5